

Data Visualization



Megan Ayers
Stat 141 | Fall 2026
Wednesday, Week 1

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Announcements/reminders

- Room change reminder for afternoon lab section
- Math/Stats colloquium tomorrow - information session about Math/Math-Stats majors

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Last Time

- Introductions
- Statistical thinking

Goals for Today

- Introduce data frames
- Motivate data visualizations
- Develop **language** to talk about the components of a graphic
- Practice deconstructing graphics
- Discuss good graphical practices

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What are/is Data?

“‘Raw data’ is an oxymoron.” – Lisa Gitelman

“Data ... is information made tractable.” – Catherine D’Ignazio and Lauren Klein

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Data Frames

Data Frames

Data in **spreadsheet**-like format where:

- Rows = Observations/cases
- Columns = Variables

ID	kind	.pred_AI	.pred_class	detector	native	name	model
1	Human	0.9999942	AI	Sapling	No	Real TOEFL	Human
2	Human	0.8281448	AI	Crossplag	No	Real TOEFL	Human
3	Human	0.0002137	Human	Crossplag	Yes	Real College Essays	Human
4	AI	0.0000000	Human	ZeroGPT	NA	Fake CS224N - GPT3	GPT3
5	AI	0.0017841	Human	OriginalityAI	NA	Fake CS224N - GPT3, PE	GPT4
6	Human	0.0001783	Human	HFOpenAI	Yes	Real CS224N	Human

- Data from **GPT Detectors Are Biased Against Non-Native English Writers**. *Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, James Zou*. **CellPress Patterns** and available in the **R** package **detectors**.

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Data Frames

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Rows = Observations/cases

What are the cases? What does each row represent?

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Data Frames

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Columns = Variables

Variables: Describe characteristics of the observations

- **Quantitative:** Numerical in nature
- **Categorical:** Values are categories
- **Identification:** Uniquely identify each case

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Data Frames

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Every time you get a new dataset, spend time exploring the variables.

Example questions:

- Is each variable capturing the desired information?
- For categorical variables, what are the categories? Do those categories adequately represent that variable?
- For quantitative variables, what values are possible? Were the data rounded or binned? Are those values actually encoding categories? What are the units of measurement?

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Types of variables

- **Identification**: Uniquely identify each case
- **Categorical**: Values represent categories
 - Usually take *non-numeric* values (e.g., Race/Ethnicity or Gender)
 - But can take numeric values! (e.g., zip code)
 - The values that a categorical variable can take are called its **levels**
 - **Ordinal** categorical variables can be ordered (e.g., level of education, variables collected with a Likert scale)
 - Categorical variables that cannot be ordered are called **nominal** (e.g., Race/Ethnicity)
- **Quantitative**: Numerical in nature
 - Those that can take a range of values are called **continuous** (e.g., age, income)
 - Those that only take particular (often whole number) values are called **discrete** (e.g., result of a dice roll)
 - Not every variable involving numbers is quantitative!

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Why construct a graph?

To **explore** the data.

To **summarize** the data.

To showcase **trends** and make **comparisons**.

To tell a compelling **story**.

Doing any of this by only looking at a data frame (even a small one) would be hard!

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Challenger

- On January 27th, 1986, engineers from Morton Thiokol recommended NASA delay launch of space shuttle *Challenger* due to cold weather.
 - Believed cold weather impacted the o-rings that held the rockets together.
 - Used 13 charts in their argument.
- After a two hour conference call, the engineer's recommendation was overruled due to lack of persuasive evidence and the launch proceeded.
- The Challenger exploded 73 seconds into launch.

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Challenger

Here's one of those charts.

BLOW BY HISTORY
SRM-15 WORST BLOW-BY
○ 2 CASE JOINTS (80°), (110°) ARE
○ MUCH WORSE VISUALLY THAN SRM-22

SRM 22 BLOW-BY
○ 2 CASE JOINTS (30-40°)
SRM-13A, 15, 16A, 18, 23A 24A
○ NOZZLE BLOW-BY

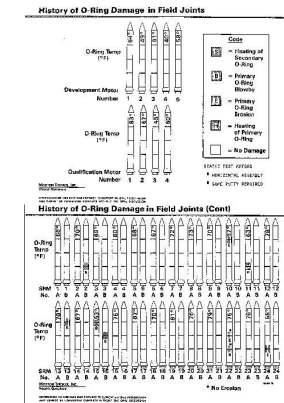
HISTORY OF O-RING TEMPERATURES (DEGREES - F)

MOTOR	MGT	AMB	O-RING	WIND
DM-1	68	36	47	10 MPH
DM-2	76	45	52	10 MPH
QM-3	72.5	40	48	10 MPH
QM-4	76	48	51	10 MPH
SRM-15	52	64	53	10 MPH
SRM-22	77	78	75	10 MPH
SRM-25	55	26	29	10 MPH
			27	25 MPH

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Challenger

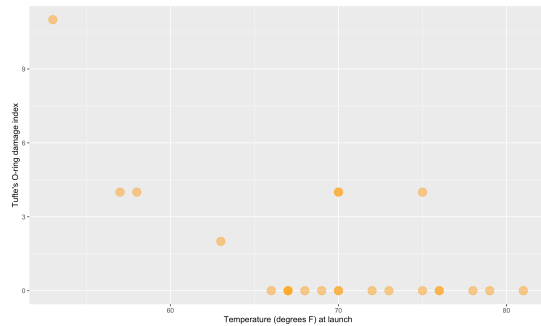
Here's another one of those charts.



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Challenger

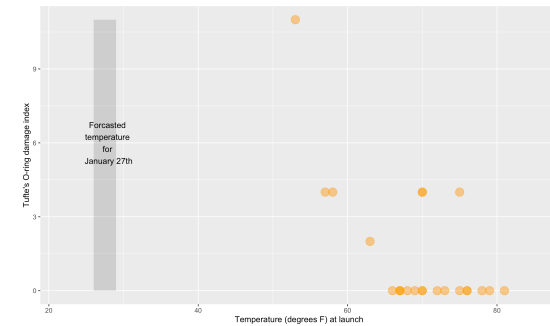
Here's a graphic created in R from Statistician Edward Tufte's data.



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Challenger

This adaptation is a recreation of Edward Tufte's graphic.



For more information on this example and other examples, check out [Tufte's book](#).

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Now let's learn the Grammar of Graphics.

We will use this grammar to:

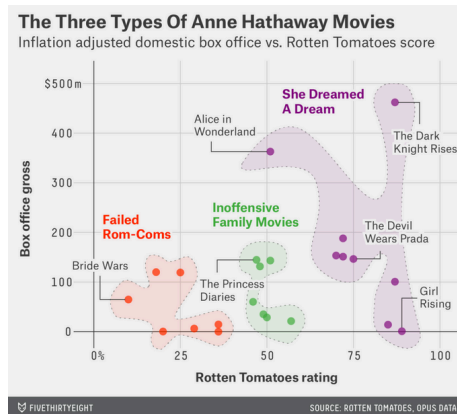
Decompose and understand existing graphs.

Create our own graphs with the R package `ggplot2`.

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Example 1: Think-pair-share

- What are the variables?
- What **geom** (i.e. shape, form) are the variables mapped to?
- What are the **aesthetics** (visual properties) of the **geom**?
- How is each variable mapped to an **aesthetic**?
- What additional context is provided? Is any missing?
- What story is the graph telling?



04:00

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Grammar of Graphics

- **data**: Data frame that contains the raw data
 - Columns are variables used in the graph
- **geom**: Geometric **form** that the data are mapped to.
 - EX: Point, line, bar, text, ...
- **aesthetic**: Visual properties of the **geom**
 - EX: X (horizontal) position, y (vertical) position, color, fill, shape
- **scale**: Controls how data are mapped to the visual values of the aesthetic.
 - EX: particular colors, log scale
- **guide**: Legend/key to help user convert visual display back to the data

For right now, we won't focus on the **names** of particular types of graphs (e.g., scatterplot) but on the **elements** of graphs.

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Example 2: Think-pair-share

- What are the variables?
- What **geom** are the variables mapped to?
- What are the **aesthetics** of the **geom**?
- How is each variable mapped to an **aesthetic**?
- What additional context is provided? Is any missing?
- What story is the graph telling?

Sexual harassment charges, by industry
Among charges filed by women, fiscal years 2005-2015

INDUSTRY	CHARGES FILED
Accommodation and food services	4,801
Retail trade	4,380
Health care and social assistance	3,898
Manufacturing	3,741
Office administration and waste management	2,350
Public administration	2,239
Professional, scientific and technical services	1,944
Transportation and warehousing	1,801
Finance and insurance	1,380
Educational services	1,340
Other services (except public administration)	1,003
Information	982
Construction	774
Wholesale trade	752
Real estate rental and leasing	611
Arts, entertainment and recreation	537
Agriculture, forestry, fishing and hunting	276
Management of companies and enterprises	213
Utilities	211
Mining	157

Not including 35,304 charges filed without a specified industry
SOURCE: EQUAL EMPLOYMENT OPPORTUNITY COMMISSION

03:00

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Visualization Considerations

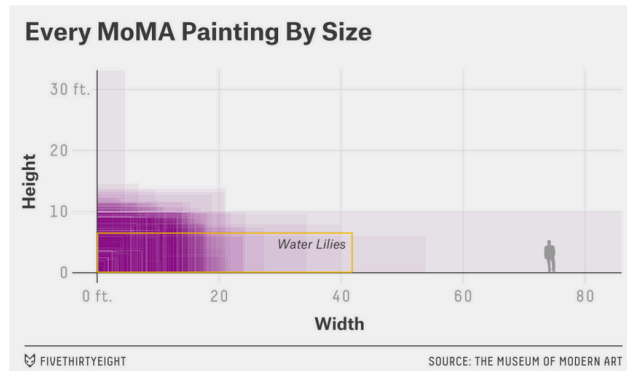
What additional context should my graphs have?

- For context, at a minimum include
 - Axis labels (with units reported).
 - Legends.
 - Data source.
- Think about the **stories/questions** your visualization answers.
- Determine what **context/background information** your viewer needs.
- Visualizing data involves **editorial choices**.
 - What to highlight.
 - What comparisons to make easy to see.
 - What scales to use.

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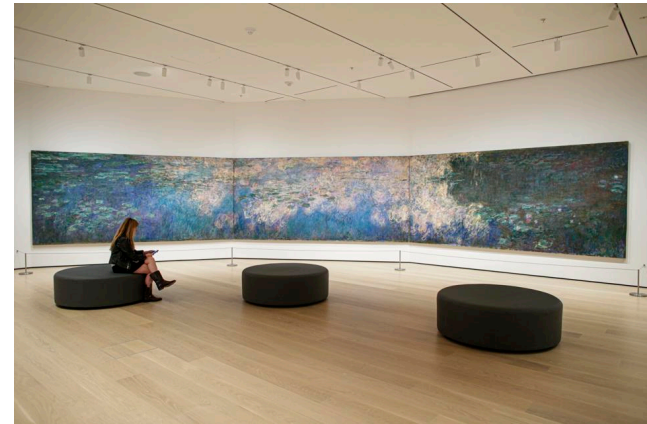
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Context Example



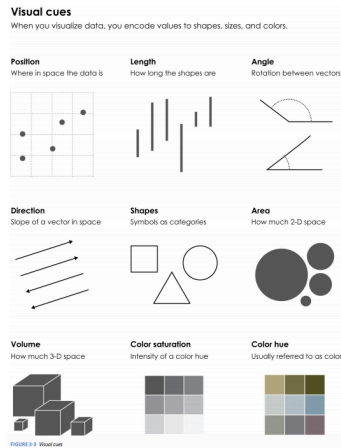
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Water Lilies



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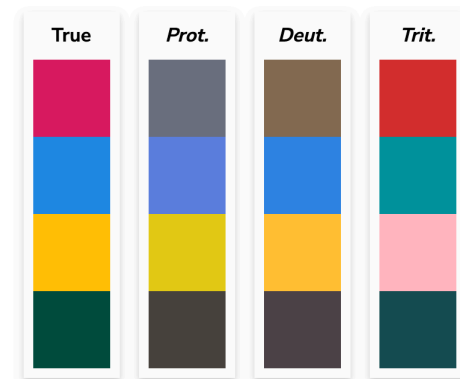
What visual cues are easier to compare?



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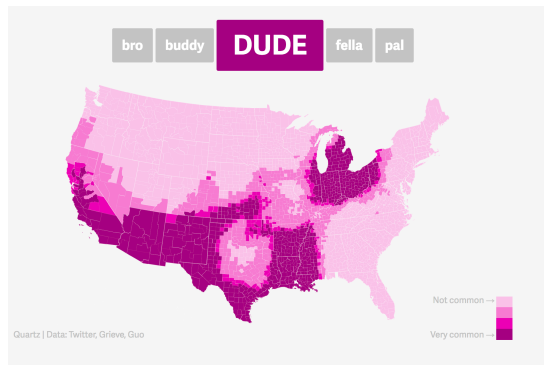
What to consider with color?

Consider color blindness (click [here](#) for a helpful tool for picking colors).



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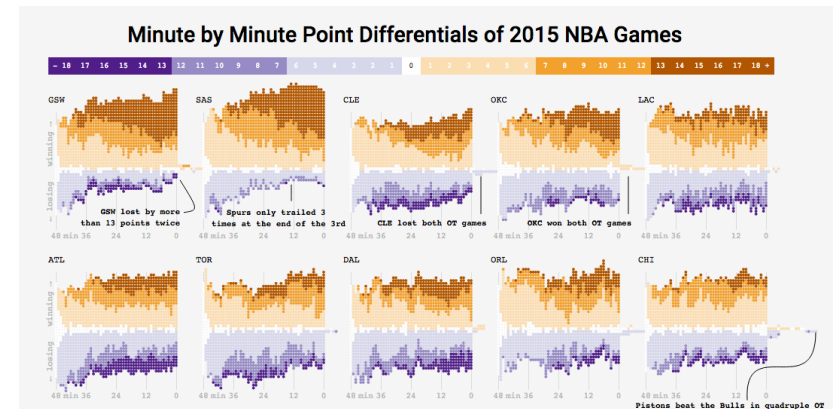
Color Palettes – Sequential



Maps, like the **Dude map** are also a great way to provide context!

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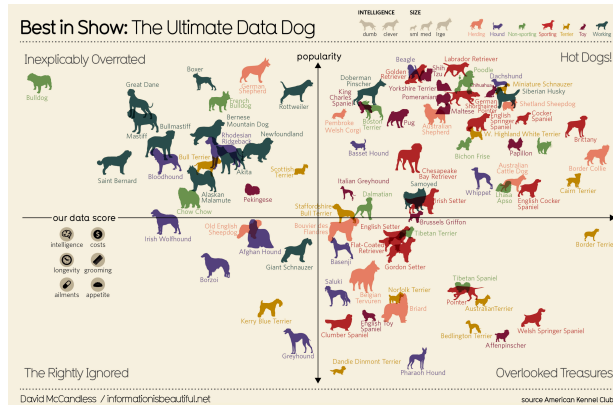
Color Palettes – Diverging



Adam Pearce's 2015 NBA Games

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Color Palettes – Qualitative

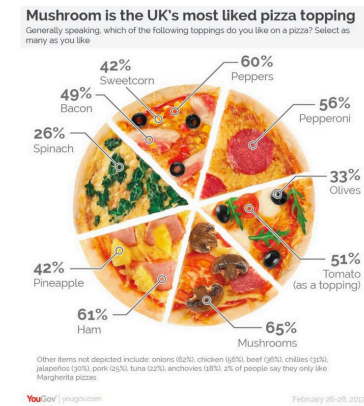


information is beautiful's Best in Show

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Bad Graphics

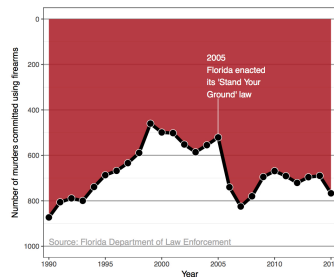
Because of all the design choices, it is much easier to make a bad graph than a good graph.



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Misleading Graphics

Be careful that your design choices don't cause your viewer to draw incorrect conclusions about the data:



- Just letting the software make all the design choices can still lead to misleading graphs (recall the Georgia COVID graph).

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Summary Thoughts on Graphical Considerations

- Good graphics are one's where the findings and insights are **obvious** to the viewer.
 - Add information and key **context**.
- Facilitate the **comparisons** that correspond to the research question.
 - Recall the three Georgia COVID counts graphs from Day 1!
- Data visualizations are **not neutral**.
- It is easier to see the differences and similarities between different types of graphics if we learn the **grammar of graphics**.
- Practicing **decomposing** graphics should make it easier for us to **compose** our own graphics.

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Recap

Today we...

- Introduced data frames
- Motivated data visualizations
- Developed **language** to talk about the components of a graphic
- Practiced deconstructing graphics
- Discussed good graphical practices

Next time

- Tomorrow in lab: intro to using RStudio to code and explore data frames
- Friday: We'll learn about the **ggplot2** package so that we can use the grammar of graphics to create beeaauutiful visualizations!!